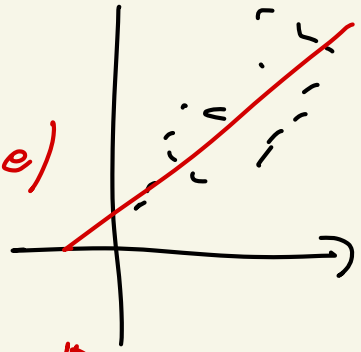



データの準備

入力: $x_i = (x_{i1}, x_{i2}, \dots, x_{id})^T \in \mathbb{R}^d$
(説明変数)

転置 (transpose)



$\underbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}_d$

出力: $y_i \in \mathbb{R}$
(目的変数)

$\mathbb{R}^2 = \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}$

データセット $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$

線形モデルの仮定

パラメタ(重み): $w = (w_1, w_2, \dots, w_d)^T$

バイアス(切片): b

予測値 $y = w^T x + b$

$$= (w_1, w_2, \dots, w_d) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} + b$$

$$= w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b$$

d 変数 $x_0 = 1$ を ~~代入~~

$$\tilde{x} = \begin{pmatrix} 1 \\ x \end{pmatrix} \leftarrow \begin{matrix} d+1 \text{次元} \\ \text{ベクトル} \end{matrix}$$

$\uparrow$
 d 次元

$$\tilde{w} = \begin{pmatrix} b \\ w \end{pmatrix}$$

$\leftarrow \mathbb{R}^{d+1}$

$$\hat{y} = \tilde{w}^T \tilde{x} \quad \leftarrow \text{見た目上 } b \text{ が消えた}$$

行列表示

7"サイン 行列

$$X = \begin{pmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_N^T \end{pmatrix} = \begin{pmatrix} 1 & x_{11}^T \\ 1 & x_{12}^T \\ \vdots & \vdots \\ 1 & x_{N1}^T \end{pmatrix}$$

← 1759

全体の予測値

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}$$

← 7"716

$$\hat{y} = Xw$$

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = \begin{pmatrix} \tilde{x}_1^T \\ \tilde{x}_2^T \\ \vdots \\ \tilde{x}_N^T \end{pmatrix} \hat{w} \equiv \begin{pmatrix} \tilde{x}_1 \hat{w} \\ \tilde{x}_2 \hat{w} \\ \vdots \\ \tilde{x}_N \hat{w} \end{pmatrix}$$

目的関数の定義

$$J(w) = \frac{1}{2} \sum_{i=1}^N (y_i - \hat{w}^T \tilde{x}_i)^2 = \frac{1}{2} \|y - Xw\|^2$$

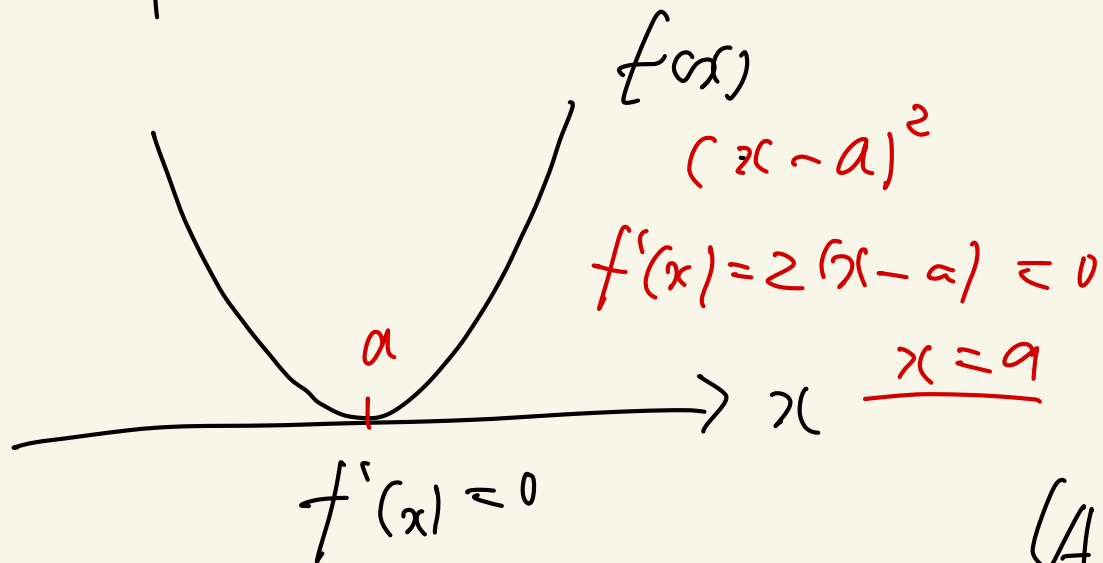
直の出力 予測値

ゴッ ← 最小にしたい。

$$2 \times 114$$

$$\| \cdot \| = \sqrt{\sum x_i^2}$$

$$\|x\|^2 = \sum x_i^2 = x^T x$$



$$(AB)^T = B^T A^T$$

$$J(w) = \frac{1}{2} (y - Xw)^T (y - Xw)$$

$$= \frac{1}{2} (y^T - w^T X^T) (y - Xw)$$

$y^T X w$
 $w^T X^T y$

(same)

$$= \frac{1}{2} (y^T y - 2 w^T X^T y + w^T X^T X w)$$

$$\frac{\partial J}{\partial w} = 0$$

ベクトルの微分

$$\frac{\partial x^T a}{\partial x} \left\{ \frac{\partial (\sum a_i x_i)}{\partial x_i} = \frac{\partial (a_1 x_1 + a_2 x_2 + \dots + a_n x_n)}{\partial x_i} \right.$$

$$\frac{\partial x^T a}{\partial x} = \begin{bmatrix} \frac{\partial (x^T a)}{\partial x_1} \\ \frac{\partial (x^T a)}{\partial x_2} \\ \vdots \\ \frac{\partial (x^T a)}{\partial x_n} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = a$$

$$\frac{\partial x^T a}{\partial x} = a$$

$$\frac{\partial a^T x c^T}{\partial x c^T} = a^T$$

$$\frac{\partial w^T A w}{\partial w} = (A + A^T) w$$

$$(fg)' = f'g + fg'$$

3. $a = A w$ $b^T = w^T A$ $c \neq c'$

$$\frac{\partial w^T A w}{\partial w} = \frac{\partial w^T a}{\partial w} + \frac{\partial b^T w}{\partial w}$$

$$= a + b = Aw + A^T w$$

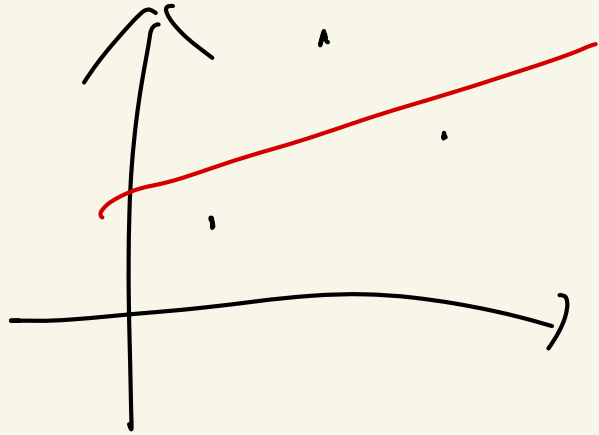
$$= (A + A^T)w$$

$$\frac{dJ}{dw} = -X^T y + X^T X w = 0$$

$$(X^T X)w = X^T y \quad (\text{正规方程})$$

~~$X^T X$~~ が "逆行列を持つ" じゃない

$$w^* = (\cancel{X^T X})^{-1} \cancel{X^T} y$$



例题 1

$$(x, y) = (1, 1), (2, 3), (3, 2)$$

$$\cancel{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$$

$$y = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$\cancel{X}^T \cancel{X} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}$$

$$\begin{pmatrix} \cancel{X} & \cancel{X} \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \frac{1}{3 \cdot 14 - 6 \cdot 6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$$

$$= \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$$

$$X^T y = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 13 \end{pmatrix}$$

$$u^* = (X^T X)^{-1} X^T y = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix} \begin{pmatrix} 6 \\ 13 \end{pmatrix}$$

$$= \frac{1}{6} \begin{pmatrix} 84 & -98 \\ -36 & +39 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$$

回帰直線

$$y = w_*^T x = \begin{pmatrix} 1 & 0.5 \end{pmatrix} \begin{pmatrix} 1 \\ x \end{pmatrix} = 1 + 0.5x$$

$$(x, y) : (1, 2) \quad (2, 3) \quad (3, 5) \quad (4, 6)$$

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix} \quad Y = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 6 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$$

$$(\cancel{X}^T \cancel{X})^{-1} = \frac{1}{4 \cdot 30 - 10 \cdot 10} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix}$$

$$= \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix}$$

$$\cancel{X}^T y = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 6 \end{pmatrix}$$

$$= \begin{pmatrix} 16 \\ 47 \end{pmatrix}$$

$$u^* = (X^T X)^{-1} \cancel{X}^T y = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix} \begin{pmatrix} 16 \\ 47 \end{pmatrix}$$

$$= \frac{1}{20} \begin{pmatrix} 480 - 470 \\ -160 + 188 \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 10 \\ 28 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5 \\ 1.4 \end{pmatrix}$$

$$y = \begin{pmatrix} 0.5 & 1.4 \end{pmatrix} \begin{pmatrix} 1 \\ x \end{pmatrix}$$

$$= 0.5 + 1.4x$$

重回歸.

$$(x_1, x_2, y) : (1, 2, 5) \quad (2, 1, 4)$$

$$(1, -1, -1)$$

$$T \bar{T}''/L$$

$$y = w_1 x_1 + w_2 x_2$$

$$\cancel{X} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{pmatrix} \quad y = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix}$$

$$\cancel{X^T} \cancel{X} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}$$

$$\cancel{X^T X}^{-1} = \frac{1}{36-9} \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$$

$$= \frac{3}{27} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$\cancel{X^T y} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 5+2-1 \\ 10+4-1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 \\ 1 & 5 \end{pmatrix}$$

$$w^* = (X^T X)^{-1} X^T y = \frac{1}{9} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix}$$

$$\Rightarrow \frac{1}{9} \begin{pmatrix} 24 & -15 \\ -12 & 30 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 9 \\ 18 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\text{Eqn } y = w_0 + w_1 x + w_2 x^2$$

$$\tau'' - A(x, y) = \begin{pmatrix} -1, 0 \\ 0, -1 \\ 1, 0 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \quad y = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

$$\begin{aligned}
 \cancel{X}^7 \cancel{X} &= \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 3 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}
 \end{aligned}$$

$$\cancel{X}^T y = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

$$(\cancel{X}^T \cancel{X})^{-1} = \frac{1}{4} \begin{pmatrix} 4 & 0 & -4 \\ 0 & 2 & 0 \\ -4 & 0 & 6 \end{pmatrix}$$

3x3の
逆行列

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0.5 & 0 \\ -1 & 0 & 1.5 \end{pmatrix}$$

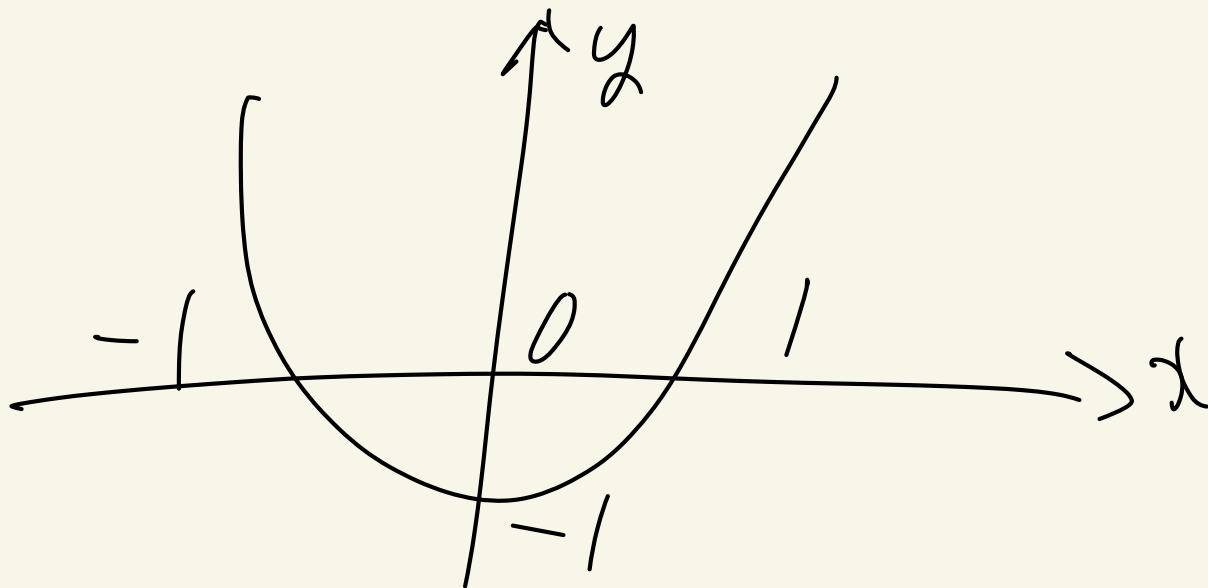
$$W^* = \begin{pmatrix} w_0 \\ w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0.5 & 0 \\ -1 & 0 & 1.5 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\hat{y} = (-1 \ 0 \ 1) \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix}$$

$$= x^2 - 1$$

$$= (x+1)(x-1)$$



$$\vec{y} = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 + 2x_2$$

$$X^T y = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 6 \end{pmatrix}$$